

ILLESHEIM AHP, Germany

WMO: 107520

Lat: 49.477N

Lon: 10.382E

Elev: 1078

StdP: 14.13

Time Zone: 1.00 (EUC)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	9.0	16.1	4.6	7.3	10.6	12.3	10.7	20.3	29.2	43.3	26.1	42.6	2.3	50	0.536

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.8	88.8	68.2	84.8	66.7	80.9	65.6	70.1	82.6	68.5	80.2	67.1	78.0	5.8	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
66.6	101.9	72.2	64.8	95.6	71.3	63.3	90.7	70.2	34.6	81.8	33.3	80.5	32.2	78.2	78.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
22.9	19.5	16.7	DB	6.3	93.9	8.0	3.4	0.6	96.4	-4.1	98.4	-8.6	100.3	-14.4	102.8
			WB	4.1	73.5	7.1	2.5	-1.0	75.3	-5.2	76.8	-9.2	78.2	-14.4	80.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	49.2	32.7	33.8	41.0	48.2	55.6	62.6	66.1	65.5	57.7	49.3	41.2	35.7
	DBStd	13.70	8.68	8.51	6.79	6.64	6.60	6.04	5.85	6.09	6.24	7.08	6.99	7.87
	HDD50	2238	535	454	285	109	23	0	0	0	9	102	276	445
	HDD65	6009	1000	873	744	503	299	117	59	69	232	488	715	910
	CDD50	1948	0	0	7	57	197	377	500	479	239	80	11	1
	CDD65	242	0	0	0	0	8	44	94	83	12	1	0	0
	CDH74	3112	0	0	0	20	130	621	1159	989	184	9	0	0
	CDH80	1036	0	0	0	5	18	199	419	356	39	0	0	0
(14) Wind	WSAvg	6.0	7.3	6.6	7.4	5.6	5.6	5.3	5.8	5.0	5.0	5.2	6.2	7.4
(15) Precipitation	PrecAvg	28.2	2.0	2.0	2.0	1.7	2.6	2.9	3.1	2.4	2.2	2.4	2.4	2.4
	PrecMax	37.6	4.4	4.8	5.6	4.6	6.2	6.1	5.9	5.4	4.9	7.9	5.1	6.7
	PrecMin	18.9	0.4	0.4	0.3	0.2	0.7	1.1	1.0	0.7	0.7	0.7	0.1	0.4
	PrecStd	4.3	1.1	1.1	1.3	1.0	1.4	1.3	1.4	1.2	1.1	1.3	1.1	1.3
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.8	58.6	67.6	76.7	83.3	90.8	94.8	93.6	84.9	75.2	62.3	54.0
		MCWB	49.4	47.1	51.8	57.6	66.0	70.5	69.2	68.9	64.6	61.2	58.6	50.1
	2%	DB	50.0	51.8	62.5	71.8	77.7	85.8	89.1	88.3	79.4	69.4	57.7	51.5
		MCWB	47.7	46.3	50.2	55.4	63.4	67.8	68.4	67.9	64.1	59.1	54.0	49.0
	5%	DB	46.6	48.5	57.5	68.0	73.7	80.9	84.5	83.8	74.8	64.5	55.0	48.8
		MCWB	44.6	44.6	47.9	54.7	59.9	66.1	66.2	67.3	62.1	57.9	52.2	46.5
	10%	DB	44.1	46.0	53.2	63.0	70.0	76.7	80.4	79.2	70.2	60.7	51.6	46.3
		MCWB	42.4	42.4	47.0	52.7	58.7	64.1	65.1	65.1	60.4	56.7	49.4	44.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	51.7	49.8	54.2	60.6	68.3	73.3	71.7	71.8	67.8	63.2	59.3	52.0
		MCDB	52.2	53.3	62.6	71.3	79.3	85.2	87.7	83.7	78.9	69.7	60.4	52.7
	2%	WB	47.9	47.4	51.6	58.1	65.3	69.4	69.5	69.9	65.1	61.0	55.3	49.6
		MCDB	49.5	50.9	59.3	68.4	74.5	81.1	82.1	82.4	75.1	66.5	57.6	51.1
	5%	WB	45.0	45.3	49.2	55.8	62.6	67.2	68.0	68.4	63.5	59.0	52.5	46.8
		MCDB	46.4	48.9	55.4	65.4	69.9	77.8	80.1	80.2	72.2	63.8	54.5	48.5
	10%	WB	42.3	42.5	47.3	53.6	60.3	65.3	66.7	66.8	61.7	56.9	49.8	44.6
		MCDB	43.8	45.4	52.2	62.5	67.9	75.0	77.5	76.9	68.8	60.3	51.4	46.0
(35) Mean Daily Temperature Range	5% DB	MDBR	9.6	13.1	18.2	21.9	21.1	21.4	21.8	22.0	20.2	16.5	11.1	9.7
		MCDBR	11.6	18.8	29.2	31.9	28.1	29.5	31.2	29.2	28.0	23.6	16.5	12.7
	5% WB	MCWBR	9.8	13.5	17.8	17.5	14.3	13.5	11.9	11.7	13.9	14.6	12.6	10.6
		MCWBR	11.4	17.3	24.2	28.6	24.2	27.0	25.4	25.5	23.4	20.4	14.6	11.6
(40) Clear-Sky Solar Irradiance	taub		0.305	0.328	0.363	0.395	0.403	0.406	0.413	0.406	0.377	0.354	0.328	0.302
		taud	2.429	2.375	2.316	2.254	2.260	2.281	2.257	2.298	2.360	2.419	2.445	2.427
	Ebn at Noon		238	258	267	269	272	271	267	263	259	245	226	221
		Edh at Noon	20	27	34	40	41	41	41	37	32	25	20	18
(44) All-Sky Solar Radiation	RadAvg		270	530	881	1350	1596	1770	1674	1453	1059	629	305	213
	RadStd		46	90	113	178	180	143	161	138	139	75	37	25

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (42 neighbors)		N/A	N/A	N/A	N/A	+0.65	N/A	N/A	N/A	N/A

Nomenclature: See separate page